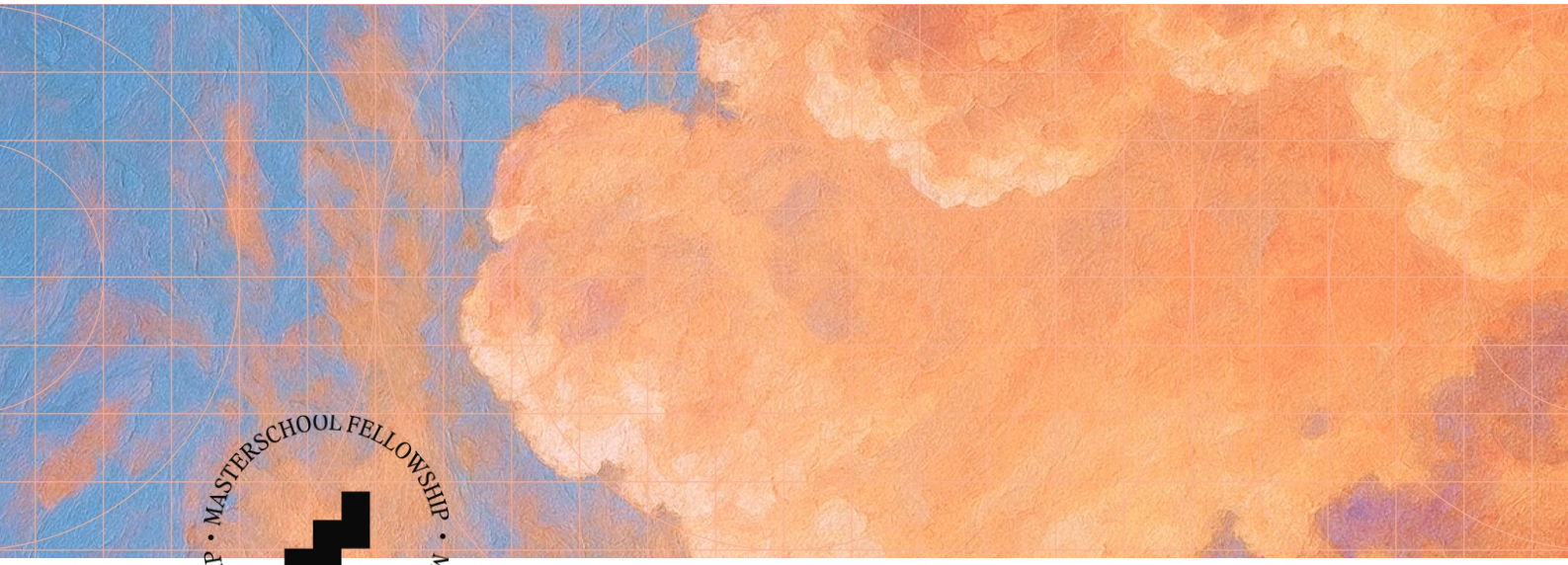

Software Engineer exercise

Building the future of education

We believe great systems unlock human potential. The tools we build are not just software - they shape how people discover new careers, new skills, and new versions of themselves.

In this home assignment, you'll build a mini version of the Masterschool admissions experience.

It's time to build, good luck 🚀

The admissions process

The admissions process is comprised of six steps, as you can see two of the steps (Interview & Sign contract) have sub-steps, called Tasks.

- Personal Details Form
- IQ Test
- Interview [contains two tasks]
  1. Schedule interview
  2. Perform interview
- Sign contract [contains two tasks]
  1. Upload identification document
  2. Sign the contract
- Payment
- Join Slack

Each user progresses through the flow, ultimately reaching one of two outcomes: "accepted" or "rejected".

Flow logic

- A **Step** in the flow is considered complete when all of its **Tasks** are finished (a step with no tasks can be considered a single task itself).
- Each task has a **condition to pass:**
  - Passed when completed (e.g., payment).
  - Passed based on results (e.g., IQ test).
- The result data for each Task is received via an incoming webhook, containing relevant information.

Step completion data

By default, a step or task is completed when its payload arrives. However, some steps or tasks have specific completion conditions. In these cases, you'll need to check the payload to determine if the user passed the step or not.

Below a list of the steps and the payload of completion for each one:

- Personal Details Form: user_id, first_name, last_name, email, timestamp
- IQ Test: user_Id, test id, score, timestamp **(passed ↔ score > 75)**
- Interview [contains two tasks]
  1. schedule interview: user_id, interview_date
  2. perform_interview: user_id, interview_date, interviewer_id, decision **(passed ↔ 'passed_interview')**
- Sign contract [contains two tasks]
  1. upload identification document: user_id, passport_number, timestamp
  2. sign contract: user_id, timestamp
- Payment: user_id, payment_id, timestamp
- Join Slack: user_id, email, timestamp

💡 **For simplicity, you don't need to store the received data in this exercise.**

Keep in mind

Our Product Managers are highly creative and frequently implement changes. Therefore, the enrollment flow should be easily modifiable, allowing for switching steps and adding or removing tasks.

Here's an example:

After running this funnel for a month, our PM noticed that the IQ Test was causing a significant drop in applicants. They decided to add a "second chance" test for users who received a "medium" score on the IQ test (between 60 and 75). This additional task should only be visible to those users—it shouldn't appear in the progress bar or step list for other applicants.

Delivery

Expose REST endpoints that allow a frontend client to:

1. POST - create a user in the system (given an email, return ID of newly created user. With this ID we should be able to work with different users in the system; you can assume it's a number or a string, but it should be unique per user)
2. GET - Retrieve the entire flow (enabling us to inform the user "You are on step 3 of 8" and display remaining steps).
3. GET - Fetch the current step and task for a specific user.
4. PUT - Mark a task as completed (including step_name, task_name, user_id, and task_payload).
5. GET - Check whether a user is accepted, rejected, or still in progress.

General information

- No actual frontend is required for this exercise.
- You may use any programming language of your choice.